

PROJECT CHARTER	Project Name: Development of Person Re-identification Algorithm for Basketball Ball-handlers in Complex Motion Scenarios
Reporting Period: 13th April - 26th April	
Team Members: Ye Li, Chengyou Long, Siyu Lei	
Overall Project Progress - Framework Evaluation: Conducted a systematic comparison between Vision-Language Pre-training (VLP) methods (CLIP-ReID) and pure vision Transformers (TransReID). Experimental results indicated that the CLIP-based cross-modal alignment could not surpass the competition baseline within the required timeframe. Consequently, the core framework was reverted to TransReID to ensure stability and high-performance output. - Mechanism Optimization: Refined the Jigsaw Patch Module (JPM) and Side Information Embedding (SIE) within the TransReID architecture, focusing on balancing local fine-grained details with global semantic consistency. - Inference Strategy Enhancement: Developed a robust post-processing pipeline, including TTA (Test-Time Augmentation) and a multi-model feature averaging tool (ensemble_test.py). Implemented independent L2-normalization for global and JPM branch features before concatenation to prevent feature scale imbalance. - Competition Materials: Completed the submission for the preliminary round of the Service Outsourcing Innovation and Entrepreneurship Competition. Responsible for drafting the technical proposal and project overview, ensuring professional presentation of the project's innovation points.	
Key Achievements - Steady growth in metrics: Through fine-grained tuning at the code level, a closed-loop improvement in metrics has been achieved in complex basketball scenarios, with mAP and Rank-1 reaching 91.5% and 94.3%, respectively. - Submission Milestone: Successfully delivered all required documentation for the Service Outsourcing competition,	
Issues and Challenges Diminishing Returns: As the model nears its performance ceiling, achieving significant mAP gains through incremental parameter tuning is becoming increasingly difficult.	
Plans for Next Biweekly Period - Failure Case Analysis: Conduct a deep dive into misidentified samples to identify specific limitations regarding occlusion and viewpoint variations. - Advanced Ensemble Expansion: Train additional model variants (different backbones or loss weights) to further leverage the potential of the ensemble_test framework.	
Risk Assessment - Risk of Overfitting: Repeated fine-tuning on specific datasets (like BallShow) may limit generalization. - Convergence Instability: Complex loss combinations (ID loss + Triplet loss) can cause instability.	